Proposal for a Swiss AI Initiative Small Project

Project Name

Scaling Laws of Knowledge Injection into Frozen Language Models

Applicants

(names, institutions, and institutional email addresses; underline scientific lead):

- Joel Barmettler, University of Zurich, joel.barmettler@uzh.ch
- Prof. Dr. Abraham Bernstein, University of Zurich, bernstein@ifi.uzh.ch *[to confirm]*
- Dr. Luca Rossetto, [institution], [email] *[to confirm]*

Request Summary

Cost Dimension	Amount
Alps Compute Hours (GPU-h)	30,000 GPU-h
Alps Storage Space (TB)	10 TB
Number of files to store simultaneously	∼500,000

Proposal Description (max. 2 A4 pages)

a. Purpose of the Swiss AI Initiative Small Project

Large language models (LLMs) hold knowledge either *in their weights* (uneditable, and unreliably recalled for long-tail entities [2]) or *as retrieved text* (faithful but costing hundreds of context tokens per query). We develop a third channel: compressing a knowledge-graph entity's neighborhood into $k \approx 8$ continuous *concept tokens* consumed natively by a *frozen* LLM — attributable (each token traces to citable Wikidata edges), editable, and ∼10× cheaper than text at equal knowledge. Our published predecessor system (WWW Companion '26, **best paper award** of its hosting workshop [1]) and extensive pilots establish that the approach works; the project's purpose is to measure the **first scaling laws of knowledge injection**: how injected-knowledge quality scales with (i) the amount of knowledge the encoder is trained on and (ii) the size of the frozen model — two axes our pilots show pull in *opposite* directions, whose unmeasured interaction decides whether this channel matters at frontier scale.

b. Current Situation of the Thematic Area

Two literatures approach the problem. *Soft-token compression* (gist tokens, ICAE, PISCO, xRAG [3, 4, 5, 6]) compresses *text* into continuous tokens; xRAG is closest to us (frozen decoder, same-model distillation) but translates a single retriever vector rather than learning compression, and evaluates neither unseen-entity generalization nor faithfulness. *Knowledge-graph injection* (GNP, GraphToken, KBLaM, Knowledge Prompts [7, 8, 9, 10]) encodes graphs per-question with labeled QA data, or precomputes per-entity vectors as lookup tables that cannot handle unseen entities. 2026 meta-analyses find current graph-token systems are not faithful carriers of structure [11]. No existing work combines an *amortized, inductive* per-entity encoder, *label-free* same-model distillation, and *causal* faithfulness evaluation — and, decisively for this call: scaling laws exist for *parametric* knowledge storage [12], but **no study measures injection quality against frozen-model scale within one model family, and no data-scaling law for a knowledge encoder exists.**

Our pilots (two consumer 24 GB GPUs, ∼60 runs, all public) de-risk the measurement: at an 8-token budget, concept tokens beat question-aware retrieval and LLM-written summaries 2.6–3.2× on the full 14,266-question PopQA benchmark, an untrained-injection control is near floor, counterfactual edge-swap probes show genuine graph reading, and injection preserves off-topic behavior. Crucially, unseen-entity accuracy scales 2.05× from 10k to 100k training entities *without saturating*, while at fixed data the injection margin *halves* from 0.8B to 2B frozen parameters: the proposal funds completing a de-risked measurement, not testing a new idea.

c. Activities

All work runs on the **Qwen3.5 family** (dense 0.8B–27B; MoE 35B-A3B/122B-A10B; every size natively multimodal), open Wikidata-derived corpora, and our public, CI-gated pipeline. Compute is justified from a *measured* anchor: one converged training run (0.6B backbone, 100k entities) takes 6.0 wall-hours on an RTX 4090 (public W&B logs); assuming a conservative 2× realized GH200 speedup and a 1.4× convergence margin, a run costs $4.2 \cdot P \cdot D$ GPU-h (P = active-parameter ratio vs. 0.8B, D = entity ratio vs. 100k). Work packages (GPU-h): **WP0** corpus scale-up to 1M entities (240); **WP1** data-scaling transect at 0.8B, k-family × 3 seeds × {300k, 1M} entities (670); **WP2** dense model transect at 100k entities, 5 sizes × k-family × 3 seeds + per-size teacher extraction, hyperparameter screens, and encoder-capacity re-sweeps (4,870); **WP3** MoE: does injection track total or active parameters? (1,730, incl. a 397B-A17B single-node-sharded stretch run); **WP4** the decisive data×model interaction corners, {300k, 1M} × {9B, 27B} (6,920); **WP5** injection-port factor — text-embedding vs. vision-token interface at 3 sizes (piloted: parity at k=8; 1,300); **WP6** community baselines: xRAG-style bridge, per-entity prefix-tuning, on-policy distillation, 2-hop rate–distortion (520); **WP7** cross-cutting evaluation: full-benchmark PopQA + EntityQuestions + counterfactual faithfulness and capability-preservation sweeps over ∼170 checkpoints (900). Subtotal 17,150; +25% contingency (restarts, cluster onboarding) = 21,440; a named stretch tier (full k-family at MoE/corners, a 3M-entity point, extra ports, convergence-verification) adds 5,570 ⇒ **request 30,000 GPU-h**; a 20k fallback preserves both core laws. The workload is ∼200 independent single-node jobs; only the two largest MoE models need intra-node sharding — and every backbone above ∼4B is memory-infeasible on our own hardware, which is the literal reason for this application. Storage: model weights, corpora, teacher caches, ∼200 checkpoints, and per-item evaluation dumps total ∼4.5 TB; with cache headroom we request 10 TB (∼500k files).

d. Team

The team carried this line of work through peer review once already: v1 [1] (Barmettler, Bernstein, Rossetto) won its hosting workshop's best paper award at The Web Conference 2026. J. Barmettler (scientific lead, PhD candidate UZH) built the v2 system end-to-end: data pipeline, training, the hardened evaluation protocol, and the pilot scaling surface. Prof. A. Bernstein (UZH) contributes long-standing expertise in knowledge graphs, the Semantic Web, and human-AI collaboration; Dr. L. Rossetto contributes expertise in multimodal retrieval and evaluation [roles to confirm]. Collaboration during the funding period: weekly experiment reviews against the pre-registered plan, shared W&B dashboards with per-item dumps for independent verification of every claim, and rotating adversarial review — the working mode that already caught and corrected four evaluation defects in the pilots. [Multi-node/Alps experience of team members to be added.]

e. Expected Outputs and Outcomes

Within six months: (1) the **scaling-law paper** — the first empirical laws for knowledge injection into frozen LLMs, with the causal-faithfulness methodology; (2) **open data**: Wikidata QA-distillation corpora at 100k/300k/1M entities with sha-manifested snapshots (CC0 sources); (3) **open weights**: all ∼200 trained encoders on HuggingFace; (4) **open source**: the full pipeline and evaluation harness (already public, 244-test CI); (5) the **open results dataset**: every run's per-item outputs, enabling third-party re-analysis without any GPU. Outcome-wise, the laws answer a go/no-go question for the field — *is* injected knowledge a channel worth scaling? — and, if the answer is yes, directly seed a **Large Project proposal**: a Swiss-made, provenance-carrying knowledge layer for open foundation models (e.g. Apertus-class models), where the Small Project's recipe, corpora, and laws determine the design points worth training at scale.

f. Importance for Switzerland, Europe, and beyond

The project advances the Initiative's trustworthy-AI mission concretely: concept tokens make model knowledge *attributable* (every token traces to citable knowledge-graph edges, and our counterfactual protocol *proves* the model uses them), *editable* (swap an edge, the answer follows), and *efficient* (∼10× fewer context tokens than retrieval at equal knowledge — a direct inference-cost and energy lever). For Switzerland and Europe, a knowledge channel that grounds open models in curated, auditable sources rather than opaque weights is a building block for digital sovereignty: public-sector and industry deployments can maintain their own knowledge graphs and update model knowledge without retraining or per-query retrieval infrastructure — with every artifact open, the value chain stays reproducible and European.

g. Ethics and Regulatory Compliance

Data: Wikidata (CC0) and public benchmarks; no personal data beyond encyclopedic facts about public entities, no human subjects, no scraping. Generated training questions come from an open-weights model and are released with their prompts; model weights used (Apache-2.0-family licenses) permit research use and redistribution of derivatives, and all released artifacts carry explicit licenses. The approach is transparency-*enhancing* (provenance-carrying knowledge aligns with EU AI Act expectations); we comply with UZH research-integrity and data-management regulations and CSCS usage policies, and publish all results, including negative ones.

Signature

By submitting this proposal, we confirm that all leading applicants have understood the requirements for participation according to the call document.

Place(s) and Date(s):

Names and signatures of the leading applicants:

References

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[12] Z. Allen-Zhu, Y. Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. arXiv:2404.05405, 2024.